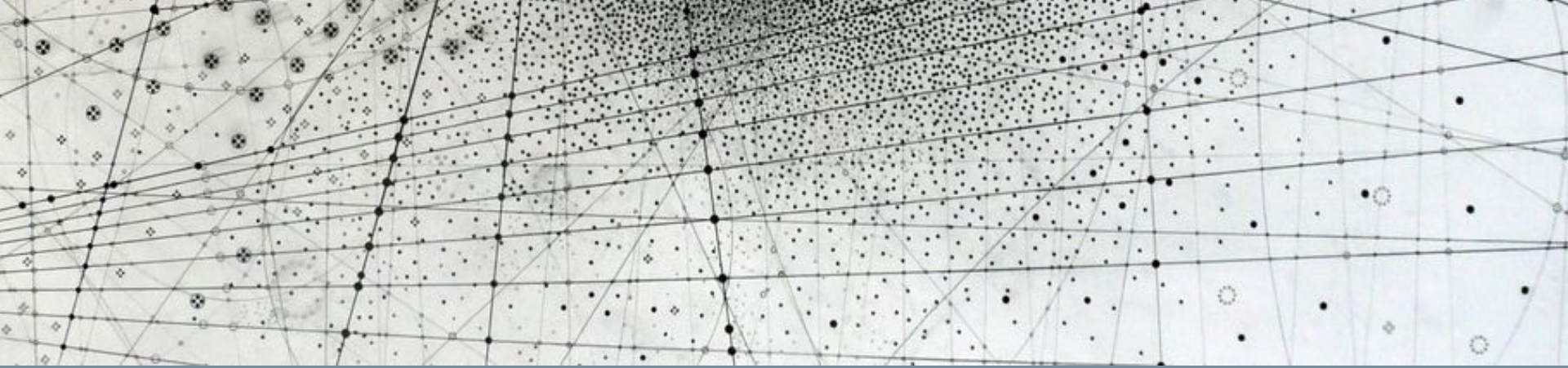


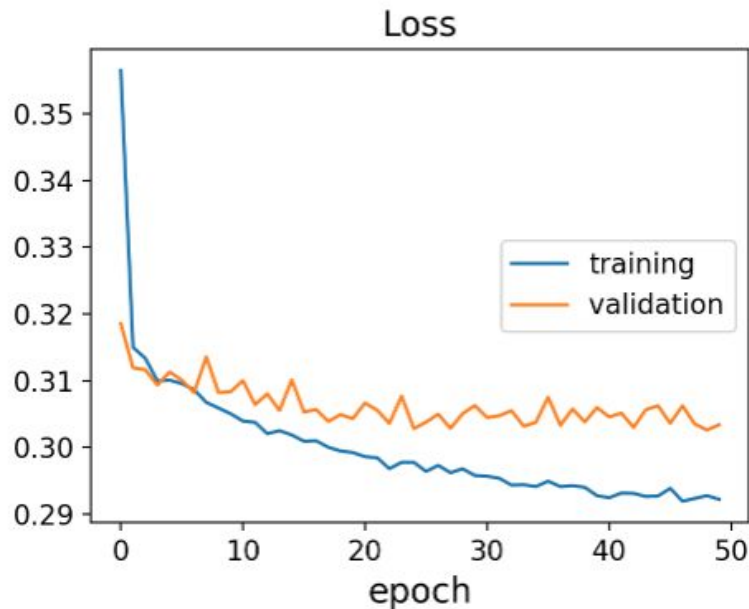


Research Methods in Machine Learning & Physics

Quick recap

Quick Recap: Regularization

- **Overtraining** occurs when your **model capacity** $\gg$ **training dataset size**
 - Training loss keeps going down (because the model is memorizing!)
 - Validation loss plateaus, or stops going down (because your model isn't generalizing well)



Regularization refers to **any intervention designed to improve generalization of your model** beyond your training data.

Quick Recap: Regularization

Examples of regularization strategies:

- Adding a term to the loss function to penalize large trainable parameters proportional to e.g. the square of the weights (**L2 regularization**)
- Randomly zero-ing out some of your weights (**dropout**)
- Stop the training when overtraining starts (**early stopping**)
- Adding new data points with noise (**data augmentation**)
- Averaging across multiple runs (**ensembling**)
- ...or maybe nothing at all! GD and SGD already have some **implicit regularizing properties**.

Quick Recap: Peer review

- **Effective peer review means:**
 - Clearly understanding your role and mission as a reviewer
 - Reading efficiently and understanding the most important details to take away
 - Assessing whether the paper's results justify/match its claims
 - Assessing if the paper is missing substantial background references or baselines
- **An example peer review structure:**
 - 1) Summarizing the paper in your own words in an uncontroversial way
 - 2) Recognizing several strengths of the paper
 - 3) Giving specific, reasonable/actionable critiques using the first person
 - 4) Stating your overall takeaways, summarizing the paper's strengths & weaknesses

Generative models, Part I: GANs & VAEs



Generative models in physics analyses

- In physics analyses, generative models can:
 - Replace a computationally-inefficient simulation pipeline
 - Interpolate into a new phase space region
 - Be used to identify anomalous events
 - Remove detector effects from data
 - Translate simulations across experimental setups
 - Help enable simulation-based inference (SBI)
 - etc.

GANs

Generative Adversarial Networks (GANs)

- Developed in 2014 by Ian Goodfellow et. al.
- A form of unsupervised learning
- Used to be state-of-the-art for image generation
- Have now fallen in popularity...
 - Training instability & mode collapse
 - Difficult to train & tune
 - Outperformed by diffusion models, in particular
- Why learn them now?
 - They are faster at inference than diffusion
 - They're still used in plenty of applications
 - The core idea was transformative in ML
 - Model-vs-model approach shows up in many other applications, too

Generative Adversarial Nets

Ian J. Goodfellow*, Jean Pouget-Abadie[†], Mehdi Mirza, Bing Xu, David Warde-Farley,
Sherjil Ozair[‡], Aaron Courville, Yoshua Bengio[§]
Département d'informatique et de recherche opérationnelle
Université de Montréal
Montréal, QC H3C 3J7

Abstract

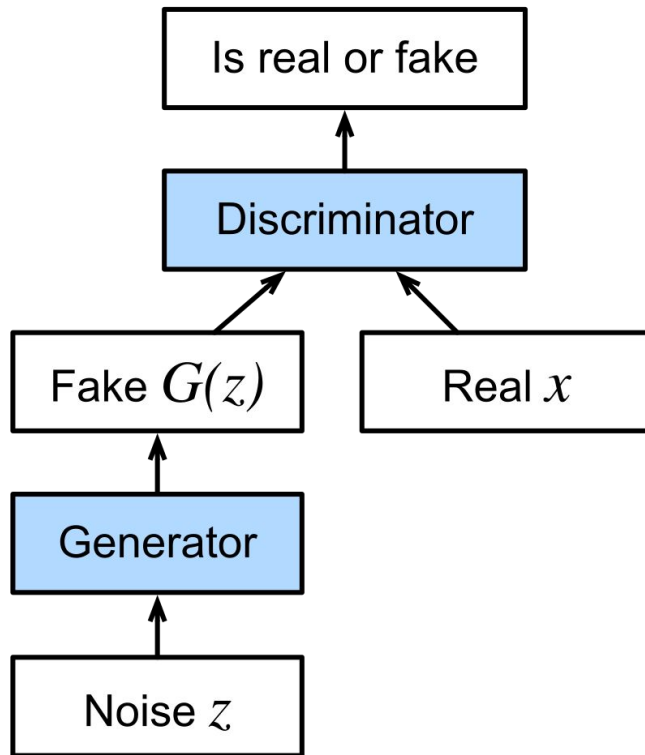
We propose a new framework for estimating generative models via an adversarial process, in which we simultaneously train two models: a generative model G that captures the data distribution, and a discriminative model D that estimates the probability that a sample came from the training data rather than G . The training procedure for G is to maximize the probability of D making a mistake. This framework corresponds to a minimax two-player game. In the space of arbitrary functions G and D , a unique solution exists, with G recovering the training data distribution and D equal to $\frac{1}{2}$ everywhere. In the case where G and D are defined by multilayer perceptrons, the entire system can be trained with backpropagation. There is no need for any Markov chains or unrolled approximate inference networks during either training or generation of samples. Experiments demonstrate the potential of the framework through qualitative and quantitative evaluation of the generated samples.

Generative Adversarial Networks (GANs)

The GAN design pits two models against each other:

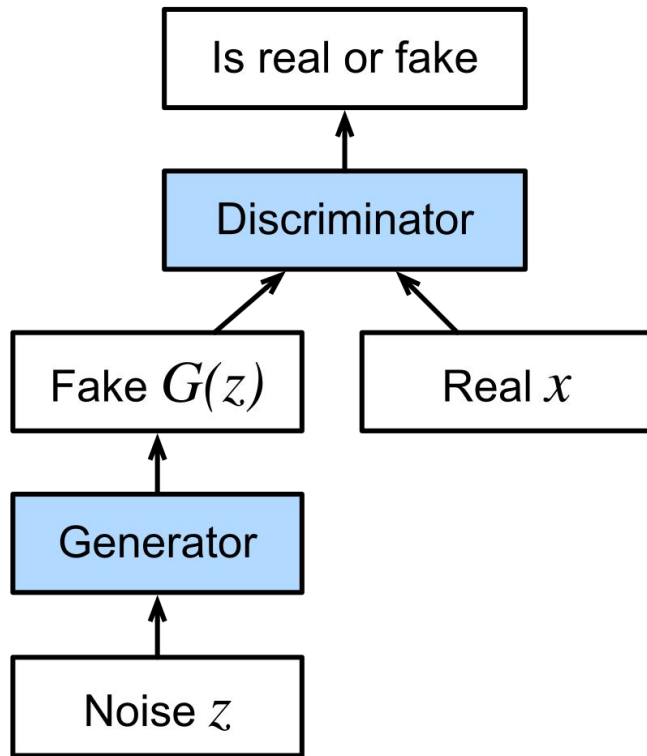
- A **generator** of “fake” outputs
- A **discriminator** between real & generated data

By iterating together, the generator improves its outputs.



Generative Adversarial Networks (GANs)

Imagine: a **forgery artist** vs. an art critic



Generative Adversarial Networks (GANs)

Real sample:



Forgery artist:



Art critic:

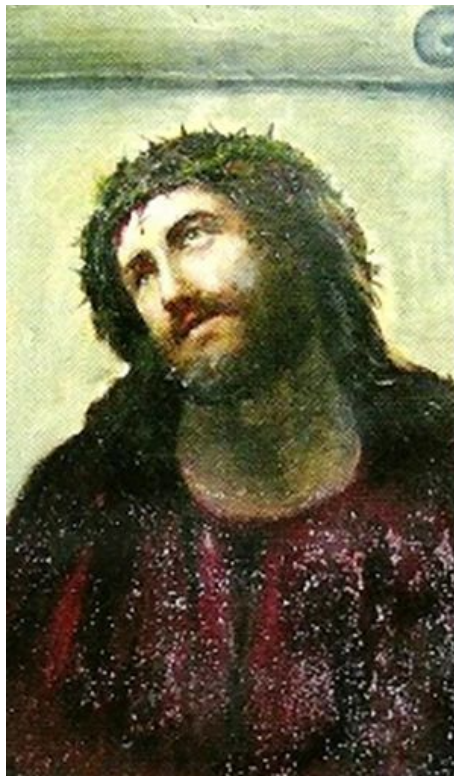
"The one on the right is fake."

Generative Adversarial Networks (GANs)

Real sample:



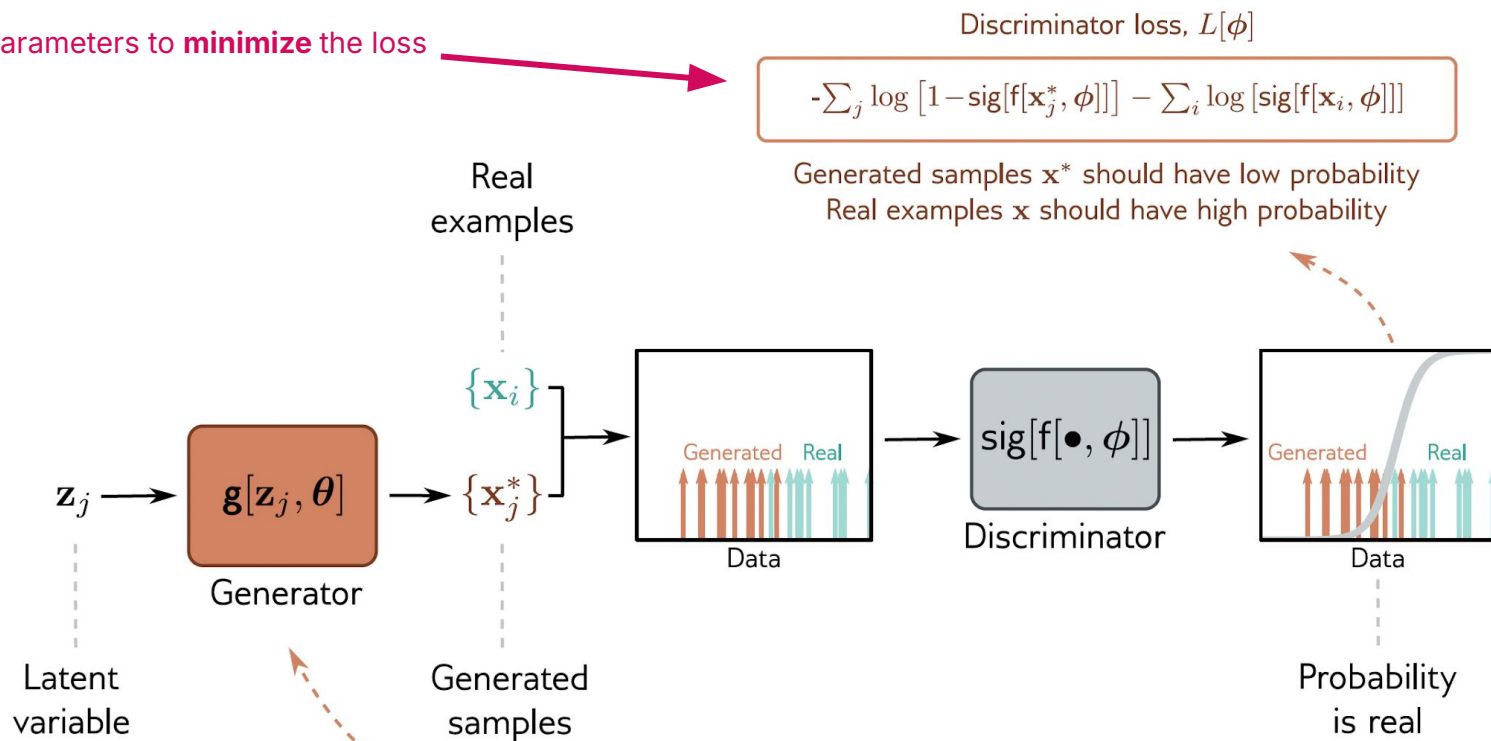
Forgery artist:



Art critic:

"I actually have no idea..."

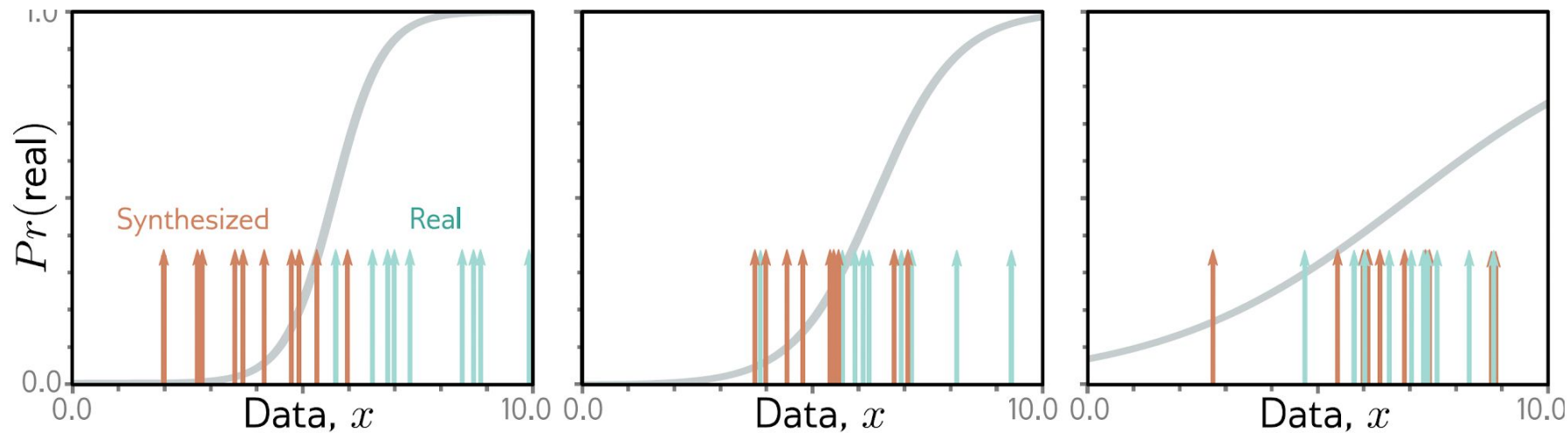
Change ϕ parameters to **minimize** the loss



Change θ parameters to **maximize** the loss

Generated samples $x^* = g[z, \theta]$
should be assigned high
probability by discriminator

Effect of a GAN over multiple iterations



VAEs

Variational Autoencoders (VAEs)

Also invented the Adam optimizer,
then co-founded OpenAI

- Introduced in 2014 by Kingma & Welling
- Like GANs, VAEs are used to generate new samples starting from latent variables z
- We already know that **autoencoders** learn a lower-dimensional representation of the input data:

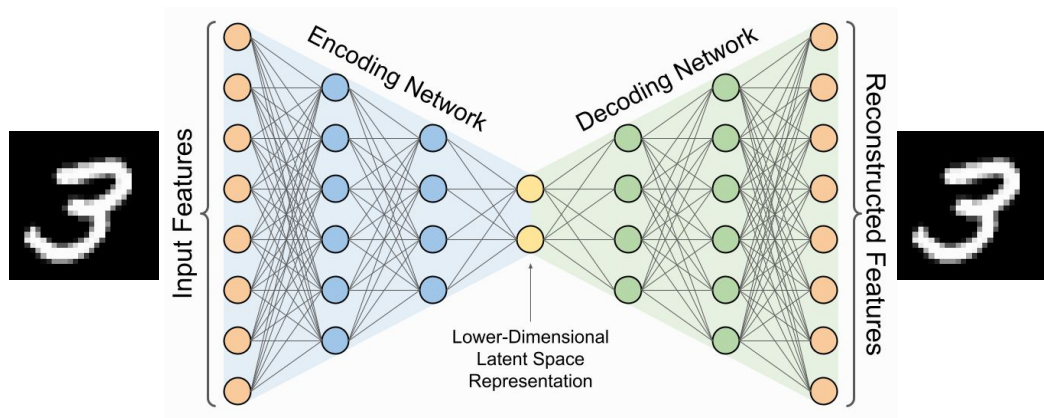
Auto-Encoding Variational Bayes

Diederik P. Kingma
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Max Welling
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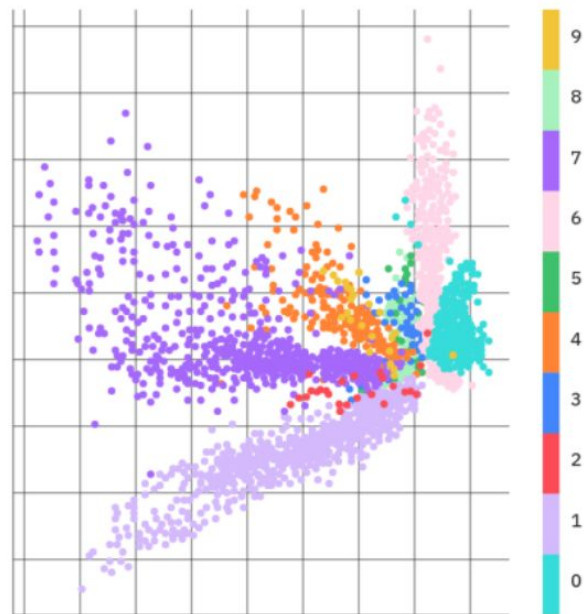
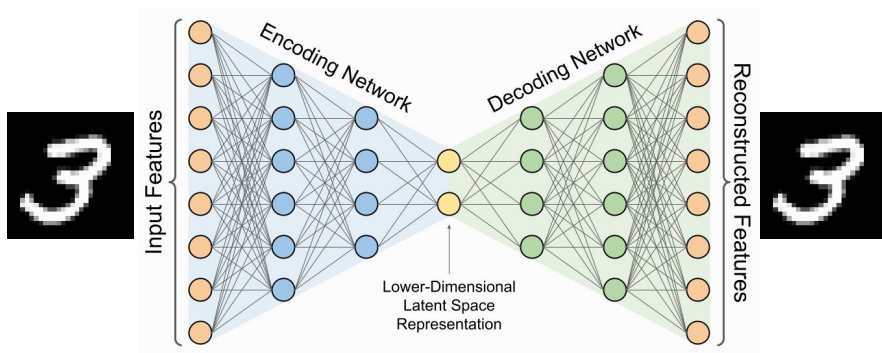
Abstract

How can we perform efficient inference and learning in directed probabilistic models, in the presence of continuous latent variables with intractable posterior distributions, and large datasets? We introduce a stochastic variational inference and learning algorithm that scales to large datasets and, under some mild differentiability conditions, even works in the intractable case. Our contributions are two-fold. First, we show that a reparameterization of the variational lower bound yields a lower bound estimator that can be straightforwardly optimized using standard stochastic gradient methods. Second, we show that for i.i.d. datasets with continuous latent variables per datapoint, posterior inference can be made especially efficient by fitting an approximate inference model (also called a recognition model) to the intractable posterior using the proposed lower bound estimator. Theoretical advantages are reflected in experimental results.



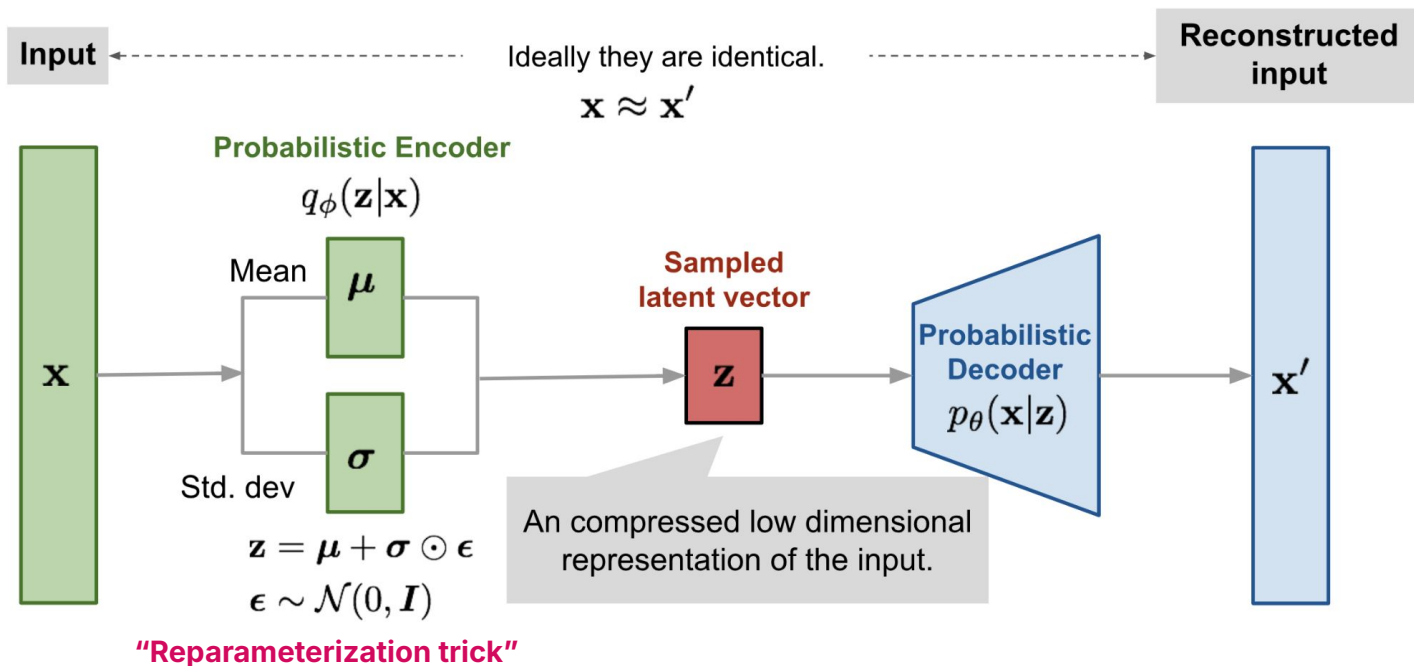
Variational Autoencoders (VAEs)

- Randomly sampling z from the latent space of a regular autoencoder (AE) will generally produce bad results, because the latent space is generally disorganized/irregular:



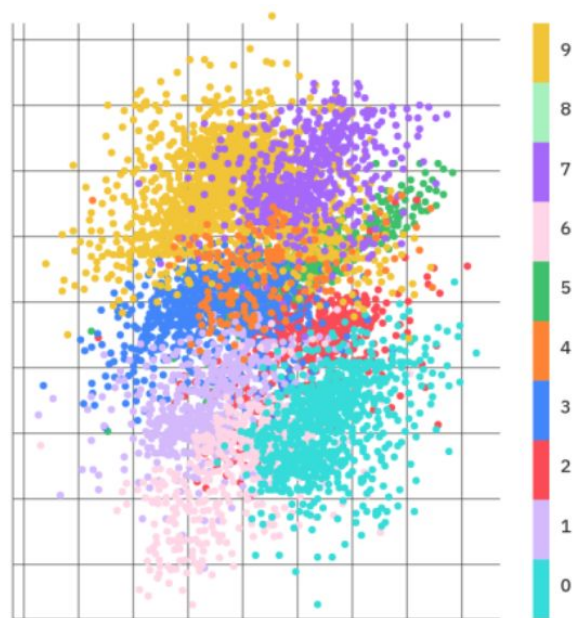
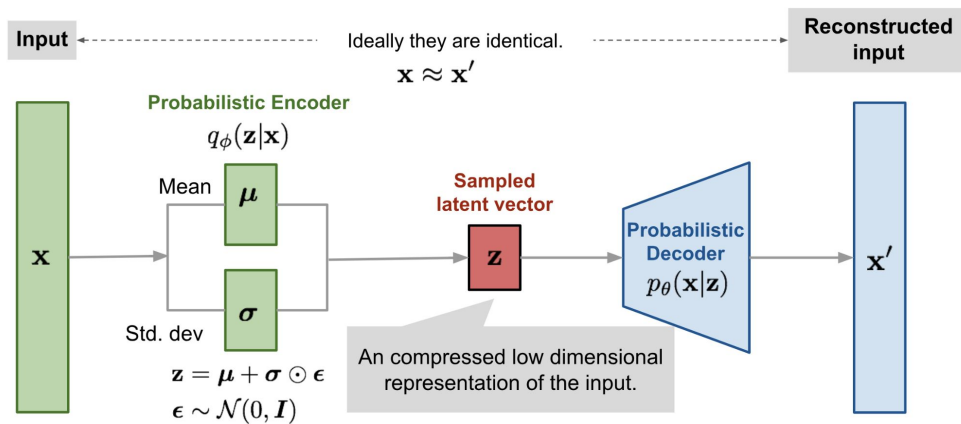
Variational Autoencoders (VAEs)

- Extends the autoencoder architecture to predict a multivariate Gaussian **distribution** instead of a static latent vector z , meaning that the latent space is much smoother and more organized:



Variational Autoencoders (VAEs)

- Extends the autoencoder architecture to predict a multivariate Gaussian **distribution** instead of a static latent vector z , meaning that the latent space is much smoother and more organized:



Variational Autoencoders (VAEs)

- The loss function combines a standard reconstruction loss with a KL divergence term:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{reconstruction}} + \beta \sum_i \text{KL}(q_i(z|x) \parallel \mathcal{N}(0, 1))$$

a.k.a. the "ELBO" = Evidence Lower Bound

- The KL divergence, a distance metric between two distributions, is defined as:

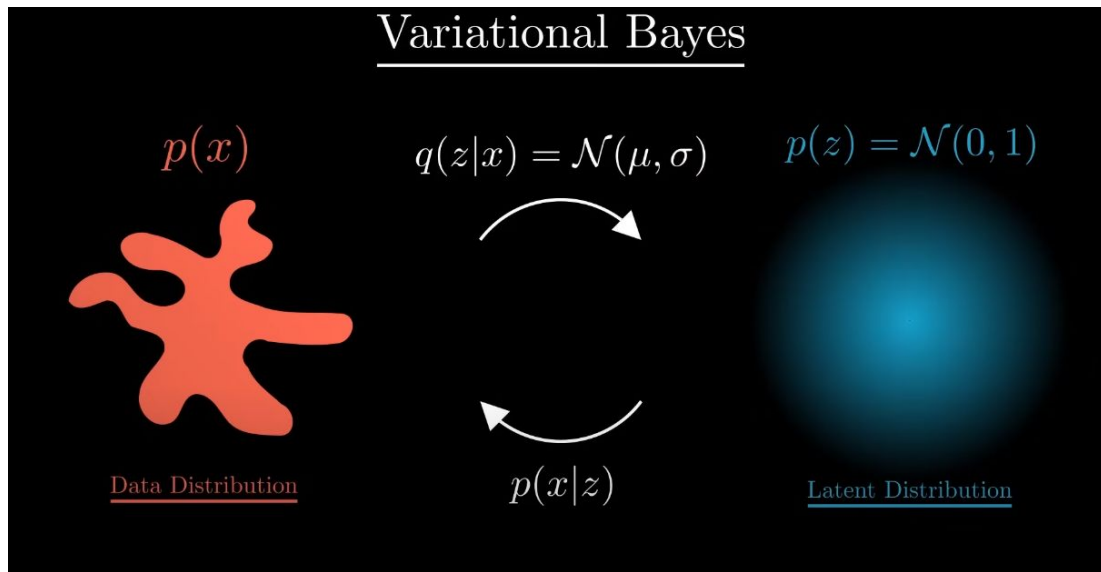
$$D_{KL}(p \parallel q) = \sum_x p(x) \log \left(\frac{p(x)}{q(x)} \right)$$

- When p and q are Gaussians, and if q is a unit Gaussian, we can write this as:

$$D_{KL}(\mathcal{N}(\vec{\mu}, \Sigma) \parallel \mathcal{N}(0, \mathbb{I})) = \frac{1}{2} \sum_i (\Sigma + \mu^2 - 1 - \ln(\Sigma))$$

Variational Autoencoders (VAEs)

- Why is it called “variational”?
 - Relates to the origins of the model in variational Bayesian statistics
 - See Understanding Deep Learning, Chapter 17, for an in-depth theoretical derivation



In-class work

Variational autoencoder notebook on the Github